

An Analysis of Factors Influencing Student Academic Performance

(COMP3125 Individual Project)

*Note: Do not used sub-title

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Abstract—Student academic performance is influenced by a wide range of personal, social, and institutional factors. This project investigates the relationship between these variables using two publicly available datasets focused on middle and high school achievement. The study combines descriptive analysis with predictive modeling to identify the strongest predictors of student performance and evaluate how behavioral, demographic, and family-related attributes contribute to academic outcomes. Findings from this analysis enhance our understanding of student success patterns and provide a foundation for data-driven strategies that support improved learning outcomes.

Keywords—student performance, educational data mining, academic achievement, predictive modeling, machine learning

I. INTRODUCTION

Understanding the factors that influence student academic performance is an important topic in education, as achievement outcomes affect long-term opportunities and overall academic success. With the growing availability of open educational datasets, it has become easier to explore patterns in student learning using real-world data. Prior work in educational data mining suggests that academic outcomes are shaped by multiple variables, including demographic characteristics, study habits, parental involvement, and school-related conditions 1. Analyses conducted on widely used datasets, such as the UCI Student Performance dataset, consistently show that factors like prior grades, attendance, and family background are closely associated with student achievement 2. These findings highlight the value of data-driven approaches for understanding how student behavior and environment relate to academic results.

This project applies those ideas by using two credible existing datasets to examine how personal, social, and school-related characteristics connect to student performance. Rather than conducting original data collection, the work focuses on exploring real datasets through descriptive analysis and predictive modeling to identify meaningful patterns. Machine learning techniques, including supervised regression and classification models, are used to evaluate whether student performance can be predicted based on variables such as prior exam scores and study habits. By integrating these analytical approaches, the project aims to provide a clearer picture of the factors that contribute to academic outcomes and illustrate how existing educational data can be used to support informed decision-making.

II. DATASETS

A. Source of dataset

This project uses two publicly available and credible datasets commonly referenced in educational data mining research. The first dataset, *Student Performance* (“student-mat.csv”), originates from the UCI Machine Learning Repository, a well-established, peer-reviewed source widely used in academic research and coursework. The dataset was originally created by Cortez and Silva in 2008 as part of a study designed to analyze secondary school performance in Portugal. Data were collected directly from school administrative records and student surveys, ensuring that the dataset represents authentic educational information. Its long-standing presence in the UCI repository and frequent use in published studies reinforce its reliability and research value.

The second dataset, *ResearchInformation3.csv*, was obtained from Mendeley Data, a reputable open-access data platform that hosts peer-reviewed and researcher-submitted datasets. This dataset was created as part of a structured academic study focused on gathering information about students’ demographic characteristics, academic behaviors, and performance indicators. It includes survey-based responses collected systematically by the original researchers, making it suitable for descriptive and correlation-based analyses. Both datasets were selected because they provide rich, multi-variable information relevant to understanding student academic performance and are sourced from platforms recognized for data transparency and credibility.

B. Character of the datasets

Both datasets in this project are provided in CSV format, which makes them easy to load and analyze using Python. The first dataset, *student-mat.csv* from the UCI Machine Learning Repository, contains 649 rows and 33 columns. These columns include a mixture of demographic information, family background, study habits, school-related details, and three important academic performance variables: G1, G2, and G3. These grade columns represent the first-period, second-period, and final grades, each recorded on a 0–20 point scale. Some features are numerical, such as absences, while others are categorical, such as parental education, school type, or whether the student has internet access. Because it includes numeric grade outcomes and many predictive features, the UCI dataset is the one used for the machine learning portion of this project.

The second dataset, *ResearchInformation3.csv* from Mendeley Data, is also organized in rows and columns but mainly contains survey-based information about student

demographics, behaviors, and academic habits. Many of its attributes are categorical (such as yes/no responses) or self-reported measures. While it is useful for descriptive analysis and understanding general trends, it does not provide the same structured numeric grade outcomes as the UCI dataset. For this reason, it is used only for descriptive and exploratory analysis, not for building predictive models.

Before analysis, I cleaned each dataset by checking for missing values, correcting inconsistent text labels, and converting categorical variables into numerical codes when necessary. No unit conversions were required because all numerical values were already in understandable formats. I also created a new binary category in the UCI dataset—such as classifying students as pass or fail based on their final grade—to support classification models. Since the two datasets have different structures and different types of variables, I analyzed them separately rather than combining them.

These preparation steps helped ensure both datasets were properly organized for descriptive work, and the UCI dataset was ready for use in machine learning.

III. METHODOLOGY

In this project, the main goal of the methodology is to build a classification model that predicts whether a student will pass or fail based on the features in the UCI Student Performance dataset. To create a classification problem, the final grade (G3) was converted into a binary label, where higher scores represent a passing outcome and lower scores represent failing. Before applying machine learning, the dataset was cleaned by encoding categorical variables, checking for missing values, and selecting important features. The data was then split into training and testing sets to fairly evaluate the performance of the models. All classification methods were implemented using the scikit-learn Python library, which provides simple, reliable tools for training and testing models. The following subsections describe the classification models used in this project.

A. Logistic Regression

Logistic Regression is a commonly used classification method that predicts the probability that a given input belongs to a particular class. The model assumes a linear relationship between the input features and the log-odds of the outcome. The core logistic function is:

$$P(\text{pass}) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n)}} \quad (1)$$

The function creates an output between 0 and 1, which is interpreted as the probability that the student passes.

Advantages:

- Simple and easy to interpret
- Fast to train
- Works well as a baseline

Disadvantages:

- Assumes linear relationships
- May not capture complex patterns

- Can be affected by highly correlated features

Logistic Regression was implemented using the `LogisticRegression` class in scikit-learn. Numerical features were scaled when needed to help with model performance.

B. Random Forest Classifier

Random Forest is an ensemble method that builds many decision trees and combines their predictions. Each tree is trained using random subsets of the data and features, which helps the model capture complex relationships while reducing overfitting.

Advantages:

- Handles both categorical and numerical features well
- Good at modeling complex, non-linear relationships
- Provides feature importance insights
- More robust than a single decision tree

Disadvantages:

- Less interpretable
- More computationally expensive
- Requires tuning for best performance

Random Forest was chosen because student success often depends on many interacting factors, and this model can handle that complexity. It was implemented using `RandomForestClassifier` from scikit-learn.

C. Decision Tree Classifier

A Decision Tree is a simple classification model that splits data into branches based on feature values. Each split tries to separate the data into more pure groups, leading to a final prediction at the leaf nodes. Decision Trees do not assume any linear relationship and can capture more flexible patterns than Logistic Regression.

Advantages:

- Very easy to visualize and interpret
- Handles categorical and numerical data
- Captures non-linear relationships
- No need for feature scaling

Disadvantages:

- Can easily overfit the training data
- Sensitive to small changes in the dataset
- Often less accurate than ensemble methods like Random Forest

Decision Trees were included because they help show the structure of the data and make it easier to understand which features are most important. They also serve as the base model for ensemble methods like Random Forest. I implemented the model using `DecisionTreeClassifier` from scikit-learn.

An excellent style manual for science writers is [7].

IV. RESULTS

In this section, present your findings using an appropriate method, such as equations, numerical summaries, or

visualizations like charts and graphs. Clearly explain all results and provide guidance on how to interpret them. If any unexpected results arise, discuss possible reasons or contributing factors. To improve clarity and organization, consider using subsections (e.g., A, B) to separate different aspects of your results.

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C. Results C

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Fig. 1. Example of a figure caption. (figure caption)

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V. DISCUSSION

Every method/project has its shortage or weakness. Please discuss the unsatisfied results in your project. And discuss the feasible suggestions of future work to revise/improve your result.

Example: xxx

VI. CONCLUSION

In this part, you should summarize your project. What important results did you find for your topic and what’s the effect of this result on the real-world?

Example: xxx

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REFERENCES

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- [1] G. Eason, B. Noble, and I. N. Sneddon, “On certain integrals of Lipschitz-Hankel type involving products of Bessel functions,” *Phil. Trans. Roy. Soc. London*, vol. A247, pp. 529–551, April 1955. (references)
- [2] J. Clerk Maxwell, *A Treatise on Electricity and Magnetism*, 3rd ed., vol. 2. Oxford: Clarendon, 1892, pp.68–73.
- [3] I. S. Jacobs and C. P. Bean, “Fine particles, thin films and exchange anisotropy,” in *Magnetism*, vol. III, G. T. Rado and H. Suhl, Eds. New York: Academic, 1963, pp. 271–350.
- [4] K. Elissa, “Title of paper if known,” unpublished.
- [5] R. Nicole, “Title of paper with only first word capitalized,” *J. Name Stand. Abbrev.*, in press.
- [6] Y. Yorozu, M. Hirano, K. Oka, and Y. Tagawa, “Electron spectroscopy studies on magneto-optical media and plastic substrate interface,” *IEEE Transl. J. Magn. Japan*, vol. 2, pp. 740–741, August 1987 [Digests 9th Annual Conf. Magnetism Japan, p. 301, 1982].
- [7] M. Young, *The Technical Writer’s Handbook*. Mill Valley, CA: University Science, 1989.

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